

Week 3: Mobility, Migration and Health

Recap: So Far

- **DTM은 실제로 관찰되는가?**

- Descriptive validity

- **사망과 출산 감소의 관계는 어떻게 설명되는가?**

- Explanatory mechanism

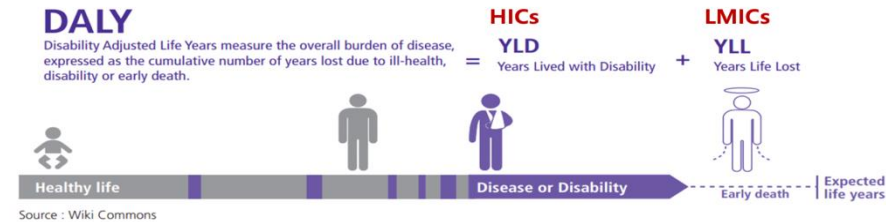
- **어떻게 사망과 출산을 줄일 수 있을까?**

- Mortality Transition (Econ vs Health)
- Fertility Transition (Econ vs Health)

- **인구구조와 인구증가관계**

- **사망의 측정 방법의 변화**

- 사망 (CDR & ASMR) – 생존 (LE/HALE) – 손실(DALY)
- 연령별 생존곡선



- **사망의 측정 방법의 변화**

- CBR – GFR - ASFR – TFR
- 연령별 출산 곡선
- Determinants: proximate determinants

- **사망감소 + 수명증가 → LMICs NCD 증가**

- **출산감소 → 아프리카?**

Migration & Mobility 측정의 어려움

- **Closed Demography**

- 인구 변화 = 출생 - 사망
- 지구 전체 인구에서는 국제이동의 영향이 없음 (우주인 제외!)

- **Open Demography**

- 인구 변화 = 출생 - 사망 + 유입 - 유출
- 이동이 추가되면서 인구 변화 과정이 복잡해짐

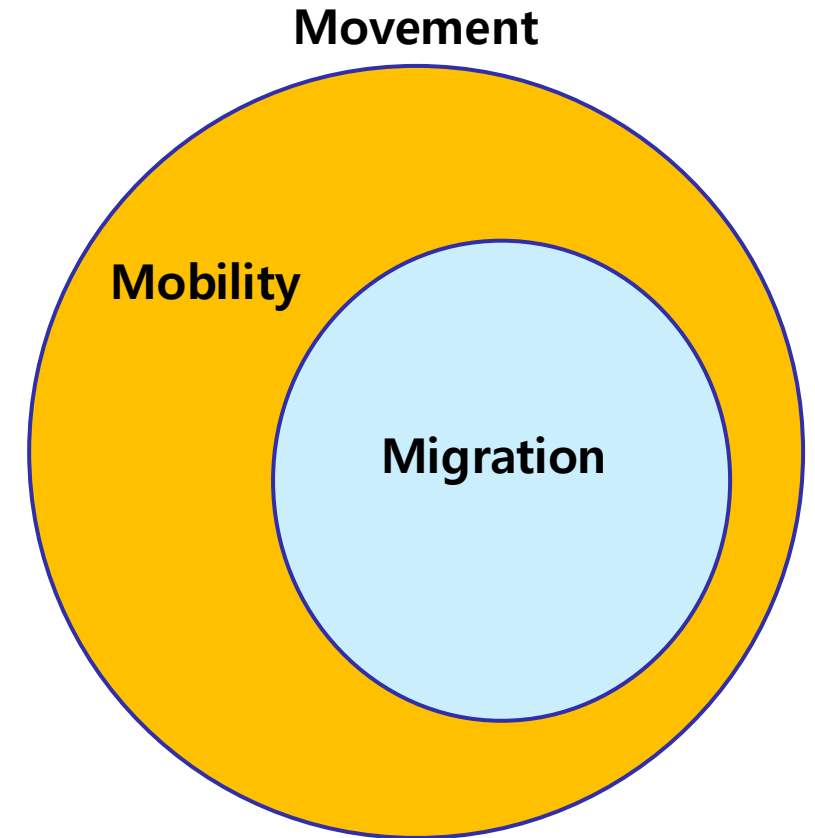
- **출산-사망 VS 이동**

출산·사망	이동
사망(한번) 출산(<~5회)	한 사람이 여러 번 이동 가능
발생 시점이 비교적 명확	이동의 시작·종료 시점이 모호

사망은 한 번, 출산은 몇 번, 이동은 수없이 발생 !

Migration vs Mobility

- **Mobility와 Migration의 공통점**
 - 모두 **출발지(Origin)** 와 **도착지(Destination)** 가 존재.
 - 사람의 공간적 이동(spatial movement)을 의미
- **Migration(이주)의 정의**
 - 사람이 한 지역에서 다른 지역으로 거주지를 옮기는 것
- **Migration의 두 가지 조건**
 - 1) **공간적 변화 (Spatial change)**
행정구역 또는 국가 간 거주지 이동
 - 2) **시간적 지속성 (Temporal duration)**
일정 기간 이상 거주 (일반적으로 12개월 이상 거주)
- **Mobility (이동)**
 - 출퇴근, 통학, 여행, 단기 출장



Migration

Today

- **Types of Migration**
- **Migration size**
- **Migration drivers**
- **Who migrates?**
- **Migration Transition**
- **Migration Health**

Types of Migration

1. Internal Migration (국내이주)

- 한 국가 내에서의 이동 (시·군·구 이동)
- In-migrant: 해당 지역으로 들어온 사람
- Out-migrant: 해당 지역에서 나간 사람

2. International Migration (국제이주)

- 국가 간 이동
- Immigrant: 입국 이민자
- Emigrant: 출국 이민자

3. Migration Flow vs Migration Stock

- Migration Stock

특정 시점에 거주하는 이주민 규모 (2025년 한국 거주 외국인 250만명)

- Migration Flow

일정 기간 동안 발생한 이동 규모 (2025년 서울 유입 인구 10만 명)

4. Gross vs Net Migration

- Gross Migration

총 이동량 (유입 + 유출)

- Net Migration

순이동량 (유입 - 유출)

Migration Size : Gravity model

Ravenstein's Laws of Migration (1885)

- 대부분의 이주는 단거리 이동
- 장거리 이동은 주로 대도시로 향함
(농촌 → 소도시 → 도시 → 대도시)
- 모든 이동에는 역이동(Counter-stream) 존재
- 경제적 요인이 주요 동인

최초의 경험적 이주 법칙

Gravity Model (Reilly & Zipf, 1940s–1950s)

- $M_{ij} = k \frac{P_i P_j}{D_{ij}^2}$ $F_g = G \frac{m_1 m_2}{r^2}$
- 인구 규모가 클수록 이주량 증가
- 거리가 멀수록 이주량 감소
- 대도시는 더 넓은 영향권을 가짐
- Distance Decay 개념 정량화

Ravenstein의 관찰을 수학적으로 설명

Gravity model : Spatial pattern

nature communications



Article

<https://doi.org/10.1038/s41467-025-56495-5>

Human mobility is well described by closed-form gravity-like models learned automatically from data

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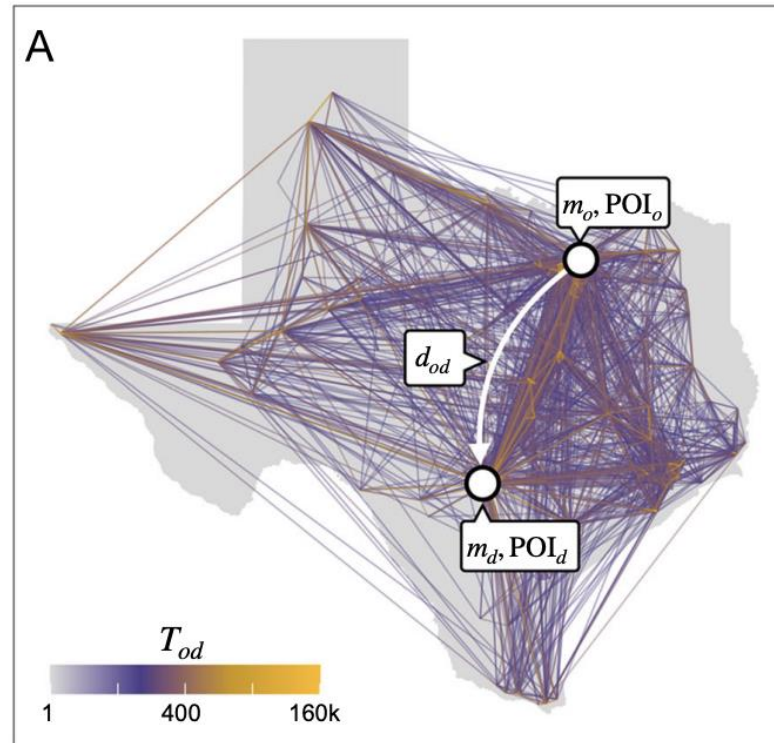
Published online: 04 February 2025

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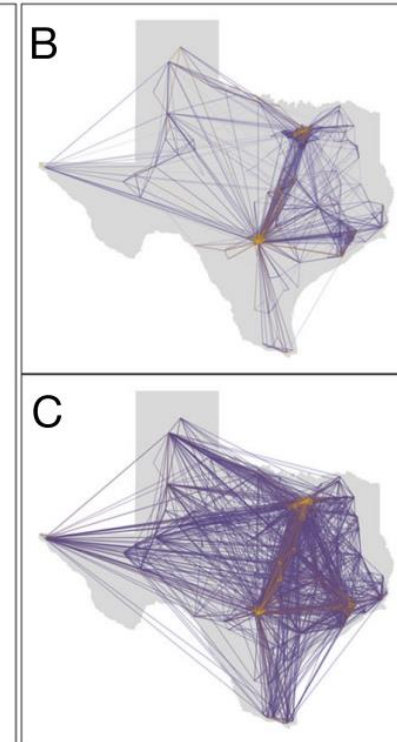
Oriol Cabanas-Tirapu¹, Lluís Danús^{1,2}, Esteban Moro^{3,4},
Marta Sales-Pardo¹✉ & Roger Guimerà^{1,5}✉

Modeling human mobility is critical to address questions in urban planning, sustainability, public health, and economic development. However, our understanding and ability to model flows between urban areas are still incomplete. At one end of the modeling spectrum we have gravity models, which are easy to interpret but provide modestly accurate predictions of flows. At the other end, we have machine learning models, with tens of features and thousands of parameters, which predict mobility more accurately than gravity models but do not provide clear insights on human behavior. Here, we show that simple machine-learned, closed-form models of mobility can predict mobility flows as accurately as complex machine learning models, and extrapolate better. Moreover, these models are simple and gravity-like, and can be interpreted similarly to standard gravity models. These models work for different datasets and at different scales, suggesting that they may capture the fundamental universal features of human mobility.

A. Observed pattern



B. Deep gravity model (39 features from origin and destination)



C. Gravity model (Only uses the population of origin and destination, distance between)

Gravity model : Temporal pattern

Humanities & Social Sciences
Communications

ARTICLE

<https://doi.org/10.1057/s41599-022-01067-x>

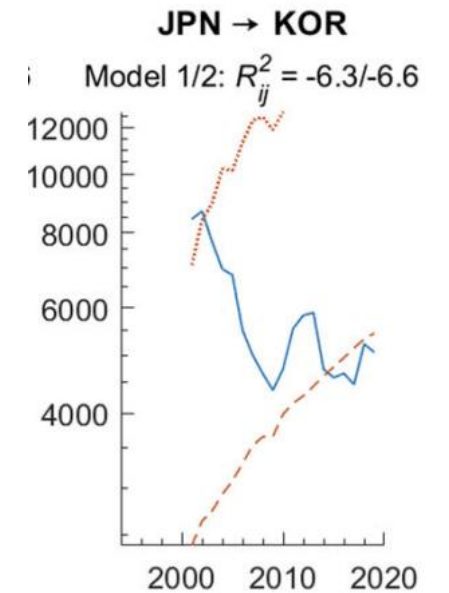
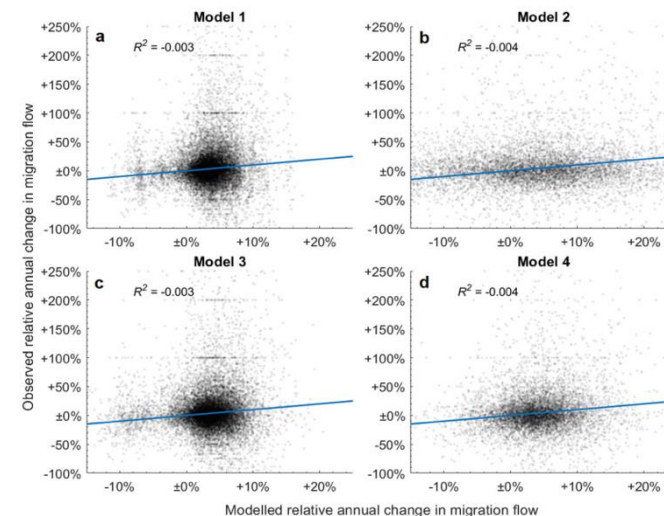
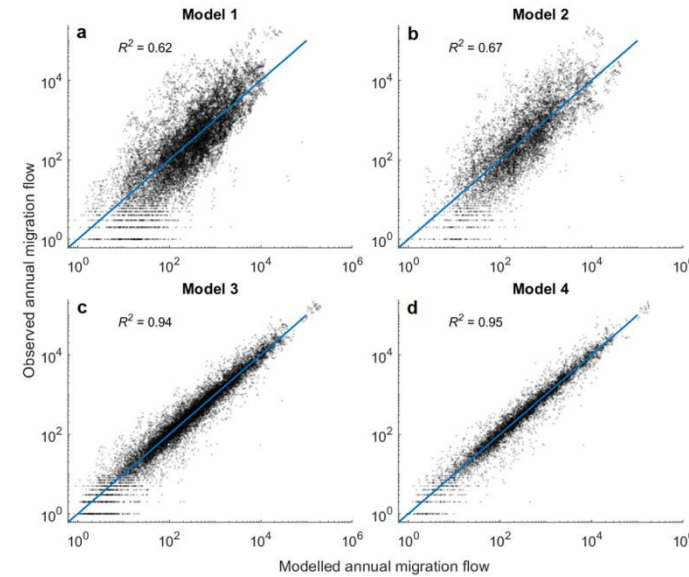
OPEN

Gravity models do not explain, and cannot predict, international migration dynamics

Robert M. Beyer¹, Jacob Schewe¹ & Hermann Lotze-Campen^{1,2}

The major social and economic impacts of international migration have led to a strong interest in better understanding the drivers of cross-border movement. Quantitative models have sought to explain global migration patterns in terms of economic, social, climatic, and other variables, and future projections of these variables are increasingly being used to forecast international migration flows. An important implicit assumption in the most widely used class of these approaches, so-called gravity models, is that their parameterisation based on panel data enables them to describe the effects of predictor variables on migration flows across both space and time, i.e., that they explain flow variation both across country pairs at a given time and across time for a given country pair. Here we show that this assumption does not hold. Whilst gravity models describe spatial patterns of international migration very well, they fail to capture even basic temporal dynamics, indeed, often worse than even the time-invariant average of the historical flows. We show that standard validation techniques have been unable to detect this important limitation of gravity models due to the different orders of magnitude of migration flows across spatial corridors, on the one hand, and over time, on the other hand. Our analysis suggests that gravity-model-based inferences about the effects that certain variables have had, or will have, on international migration over time may in reality

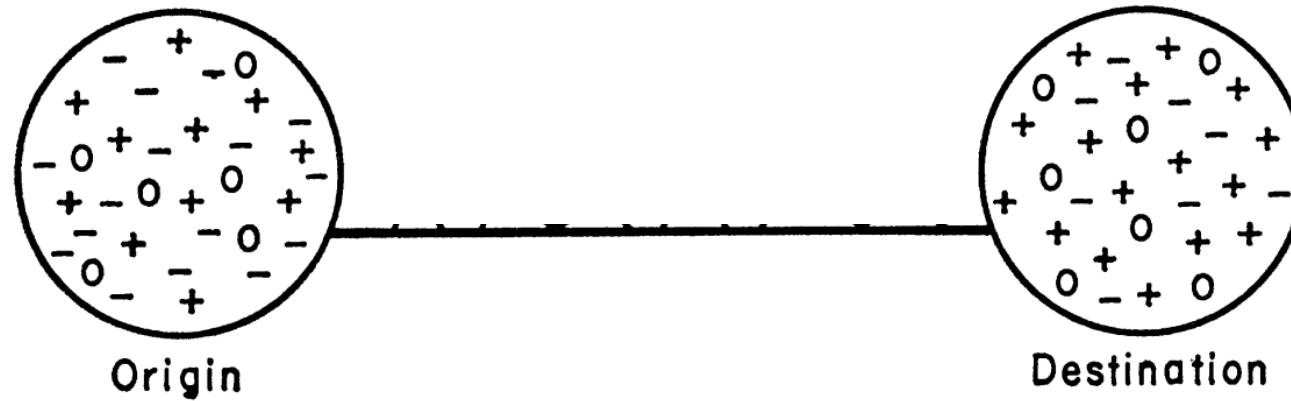
Beyer, R. M., Schewe, J., & Lotze-Campen, H. (2022). Gravity models do not explain, and cannot predict, international migration dynamics. *Humanities and Social Sciences Communications*, 9(1), 1-10.



Gravity Model : Strengths and Limitations

Strengths	Limitations
어느 지역 간 이동량 이 큰지 예측 가능	왜 사람들이 이동 하는지는 설명하지 못함
인구 규모와 거리의 효과 설명	정책 변화의 영향 반영 어려움
도시 간 이동의 공간적 패턴 설명	난민 위기, 전쟁, 코로나 같은 충격 설명 어려움
현재의 이동량(Migration flow) 예측 가능	미래의 이동 변화 예측 한계

Migration Drivers : Push – Pull Framework



1. Push Factors (출발지)

사람을 떠나게 만드는 요인

- 실업, 빈곤
- 전쟁, 정치불안, 교육 서비스부족

예시) 농촌 → 도시 이동

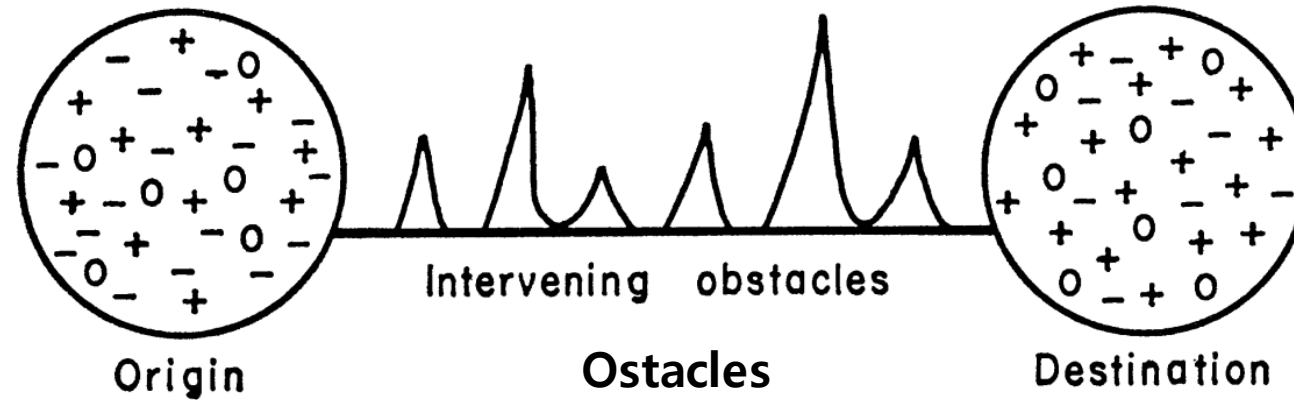
2. Pull Factors (도착지)

사람을 끌어당기는 요인

- 일자리, 높은 임금
- 교육 기회, 의료서비스

예시) 서울, 미국

Migration Drivers : Push – Pull Framework



1. Push Factors (출발지)

사람을 떠나게 만드는 요인

- 실업, 빈곤
- 전쟁, 정치불안, 교육 서비스부족

예시) 농촌 → 도시 이동

- 비용
- 비자·이민정책
- 연령

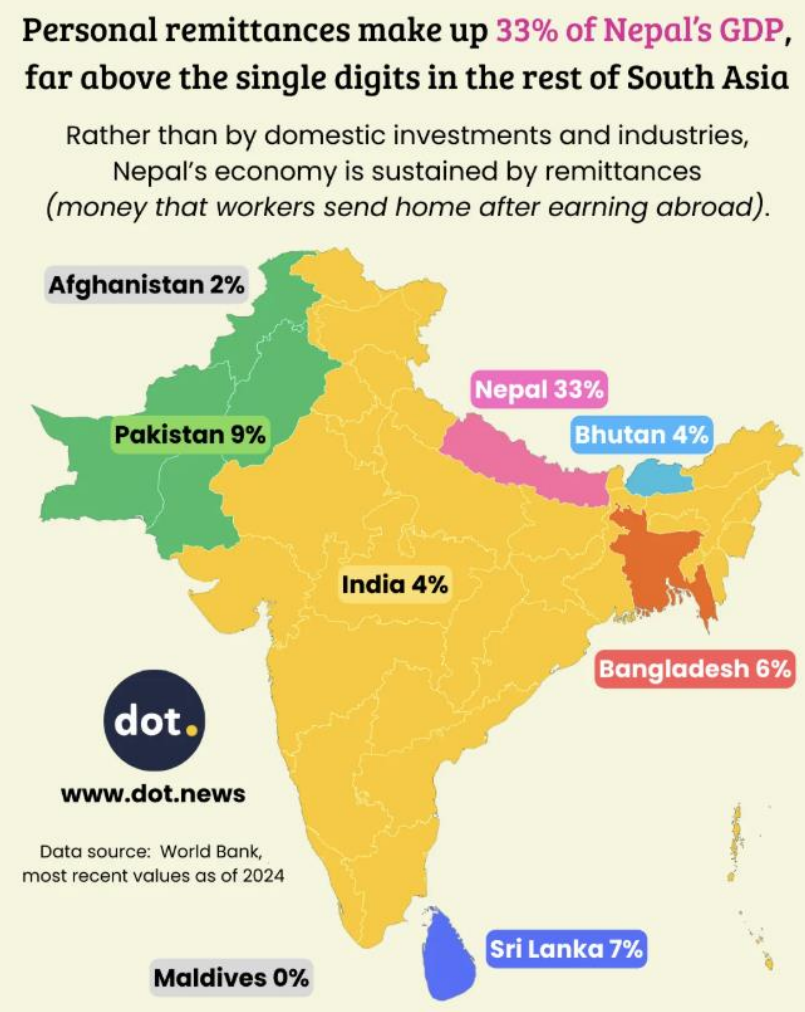
2. Pull Factors (도착지)

사람을 끌어당기는 요인

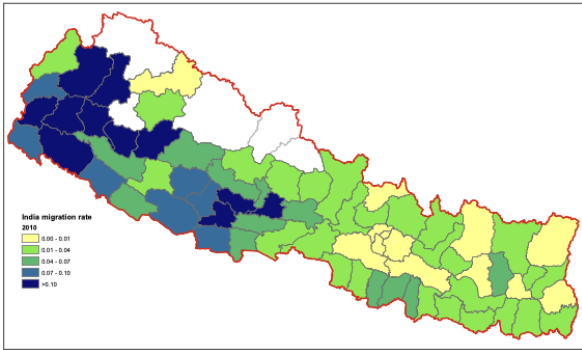
- 일자리, 높은 임금
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예시) 서울, 미국

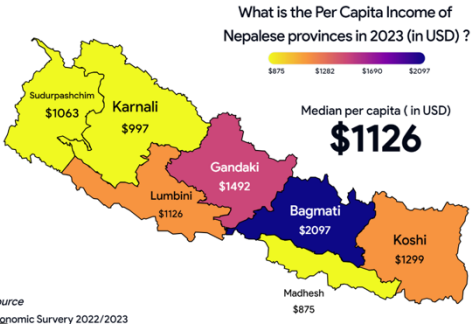
Migration Drivers : Nepal Case



Origin (Nepal)	Factor	Destin ation	Mig	Interpretation
강수량 증가 → 농업소득 증가	Push	India	↑	이주 비용을 감당할 수 있게 됨
강수량 증가 → 농업소득 증가	Push	Gulf	-	소득 증가만으로는 높은 이주 비용을 감당하기 어려움
내전 및 정치적 불안	Push	India	↑	안전한 지역으로 이동
내전 및 정치적 불안	Push	Gulf	↑	갈등 회피 및 소득 기회 추구



네팔 → 인도 이주율



지역별 GDP

Shrestha, M. (2017). Push and pull: A study of international migration from Nepal. *World Bank Policy Research Working Paper*, (7965).a

Who Migrates : Age pattern of migration

α_1 = rate of descent of pre-labor force component

x_l = low point

 λ_2 = rate of ascent of labor force component

x_h = high peak

 α_2 = rate of descent of labor force component

x_f = retirement peak

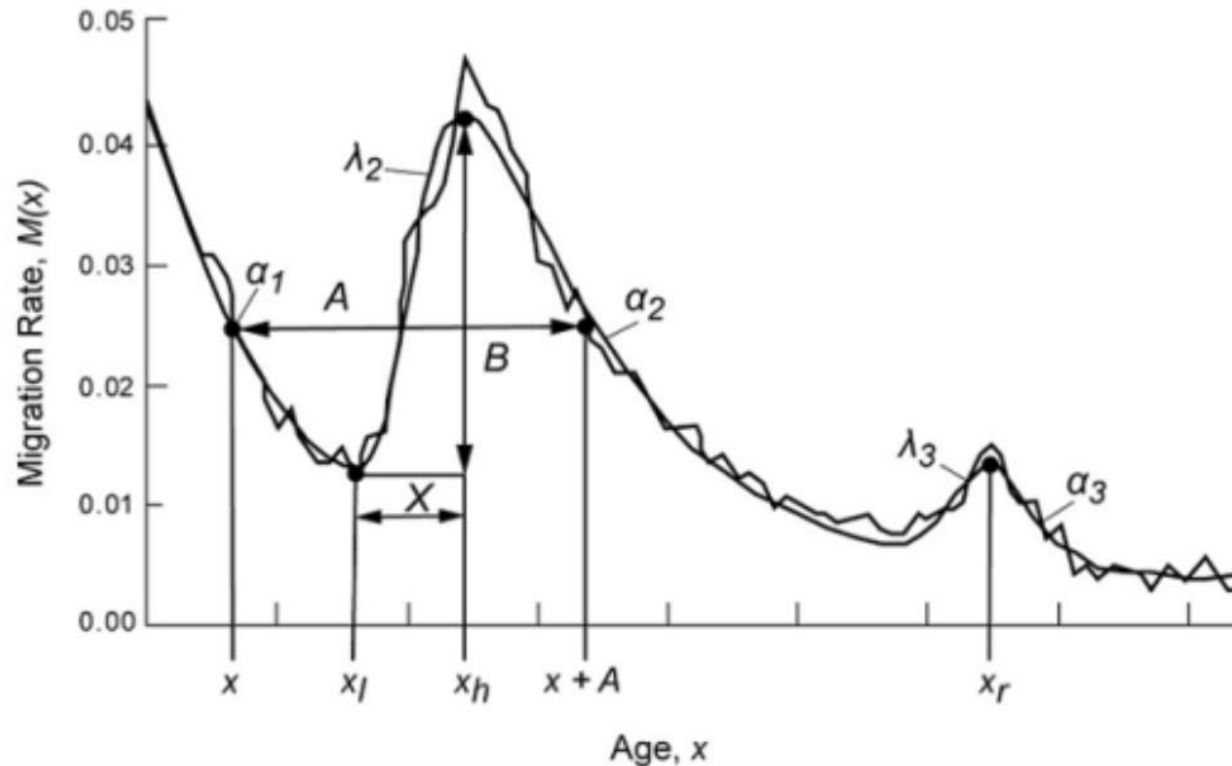
 λ_3 = rate of ascent of post-labor force component

X = labor force shift

α_3 = rate of descent of post-labor force component

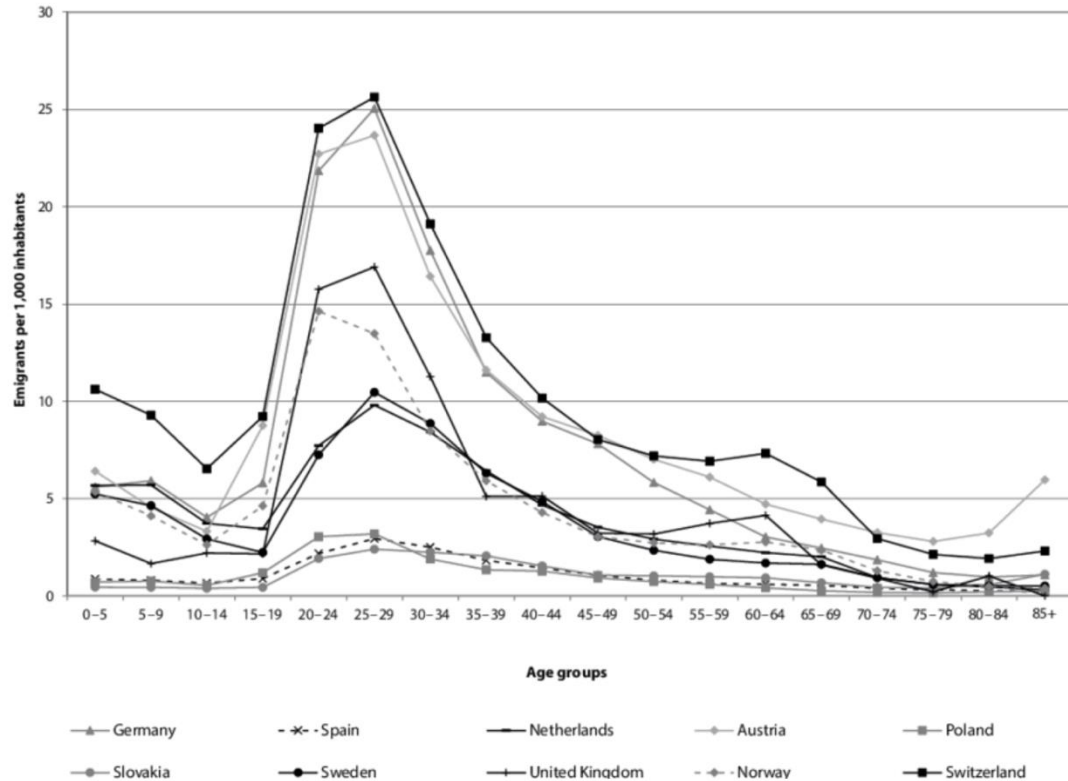
A = parental shift

$B = \text{jump}$

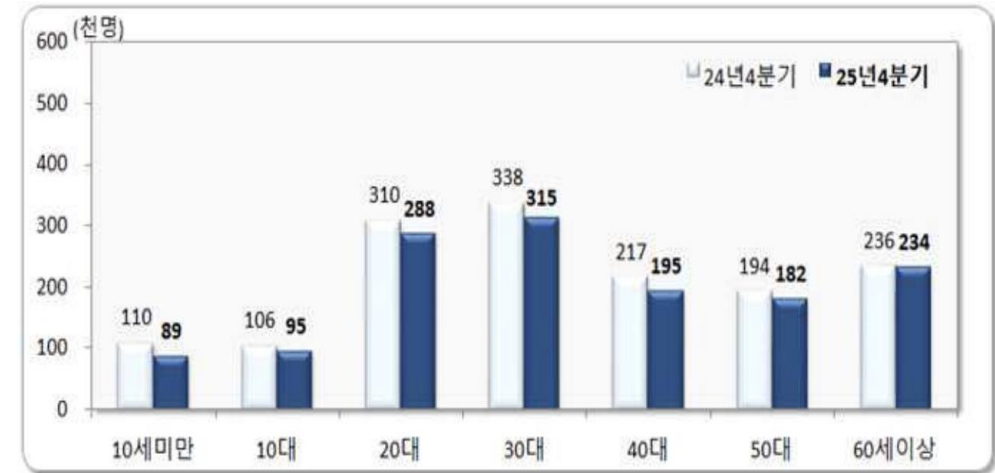


Age	Main Reason for Migration
0-14	부모를 따라 이동 (Family migration)
15-24	진학, 취업 준비, 군복무, 독립
25-34	취업, 결혼, 주택 마련
35-49	직장 이동, 자녀 교육
50-64	이동 감소 (직업·가족 안정)
65+	은퇴, 건강·돌봄 목적 이동

Who Migrates : Europe and Korea



[그림 3] 2025년 4/4분기 연령별 인구이동



2025년 12월 국내인구이동통계

Migration Transition

How does mobility change with development? (Zelinsky, 1971)

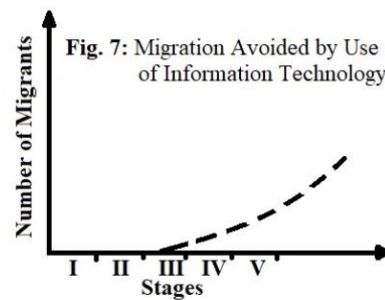
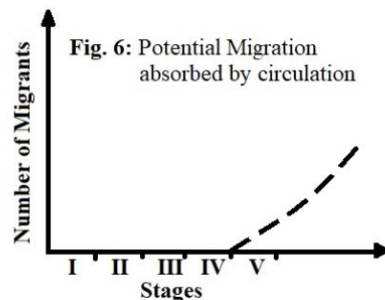
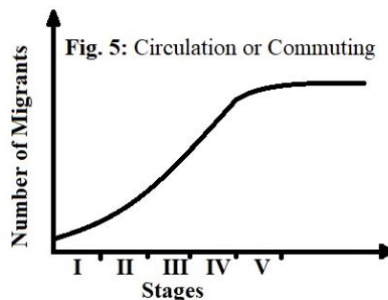
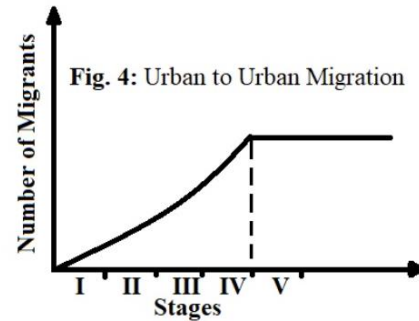
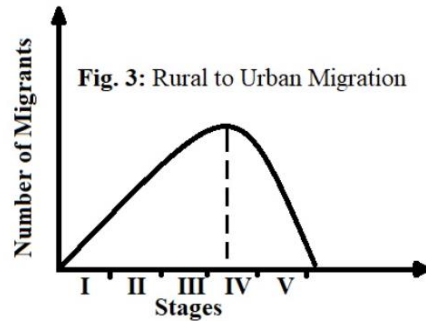
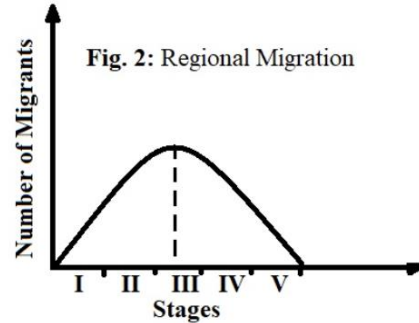
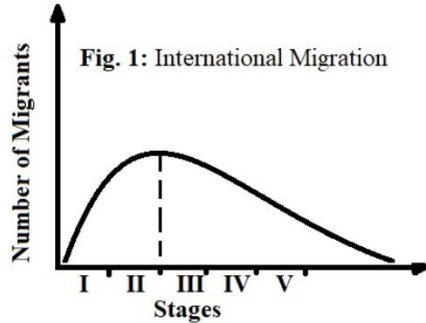


Fig 1. International Migration (국제이주)

초기 산업화(Stage II~III)에서 급증 새로운 정착지와 일자리 탐색

Fig 2. Regional Migration (국내 장거리 이동)

국가 내부의 개척지로 이동 산업화 시기에 급증

Fig 3. Rural → Urban Migration (농촌→도시)

산업화와 도시화의 핵심 이동 Stage III~IV에서 최고점

Fig 4. Urban → Urban Migration (도시→도시)

사회가 발전할수록 증가 대도시 간 이동이 활발

Fig 5. Circulation / Commuting (통근·순환이동)

교통 발달과 함께 지속 증가

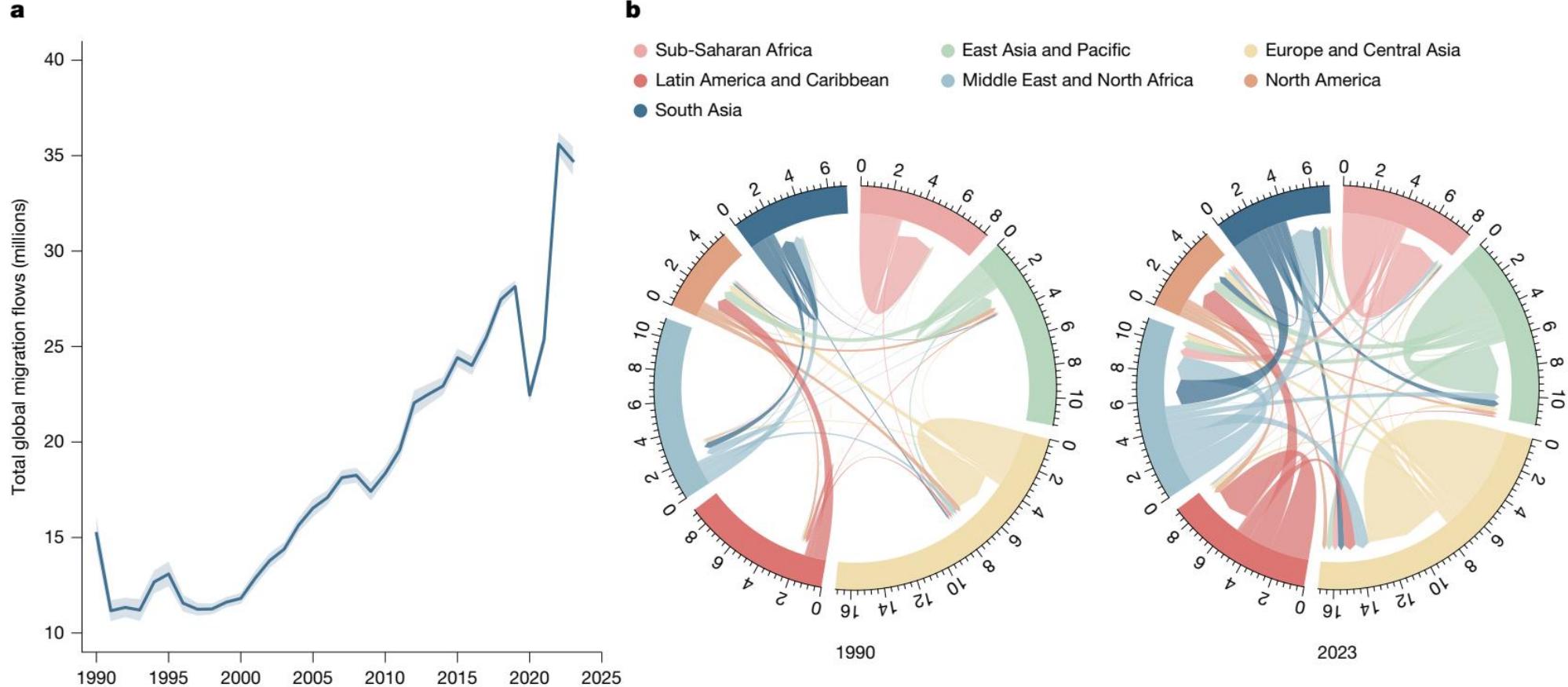
Fig 6. Migration Absorbed by Circulation

원래는 이사(Migration)해야 했던 사람들이 통근(Commuting)으로 대체

Fig 7. Migration Avoided by Information Technology

정보통신기술 발달로 이동 자체가 불필요

Migration Transition



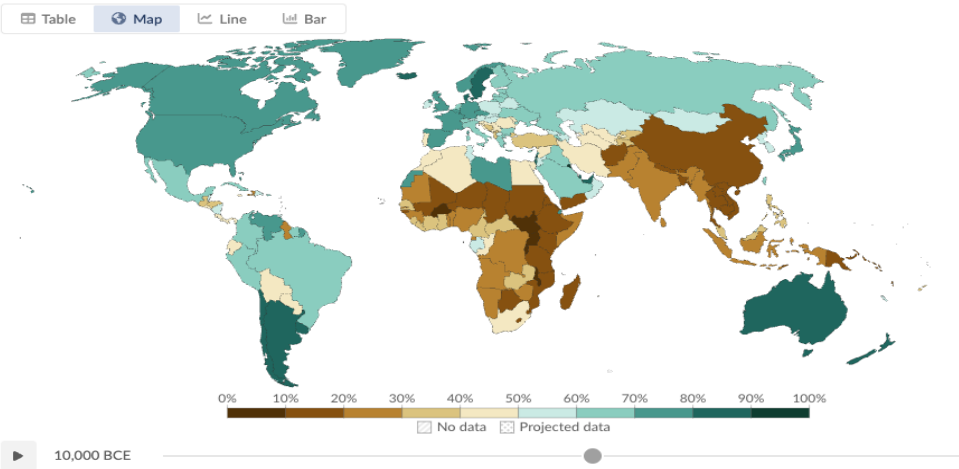
- 유럽·북미 중심 국제이주

- 아시아·중동 중심 국제이주 확대
- 새로운 경제권 등장
- 노동시장 세계화

Urban Transition

Share of population living in urban areas projected to 2050, 1980

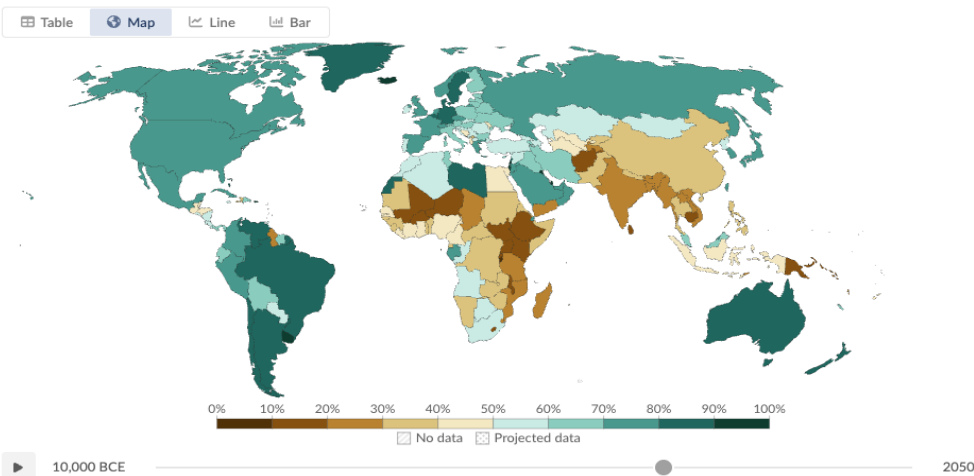
Share of the population living in urban and rural areas, combining UN estimates and projections with modelled historical data back to 10,000 BCE.



Our World
in Data

Share of population living in urban areas projected to 2050, 2000

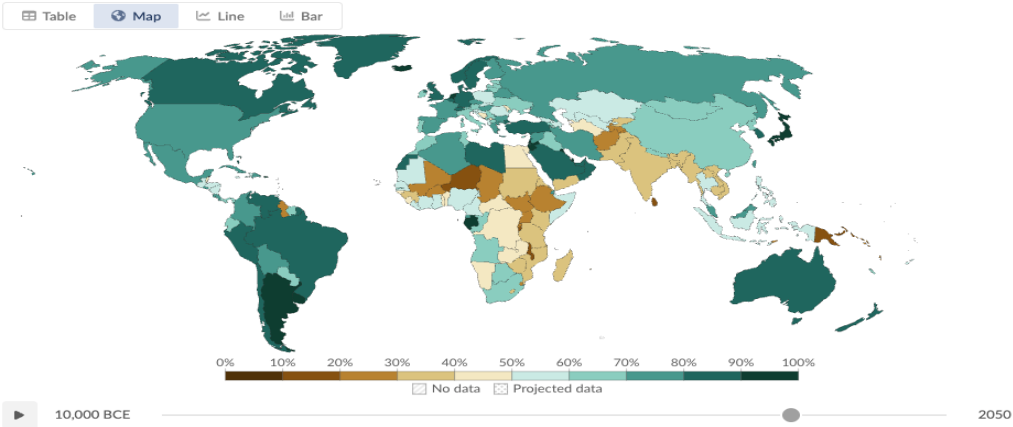
Share of the population living in urban and rural areas, combining UN estimates and projections with modelled historical data back to 10,000 BCE.



Our World
in Data

Share of population living in urban areas projected to 2050, 2020

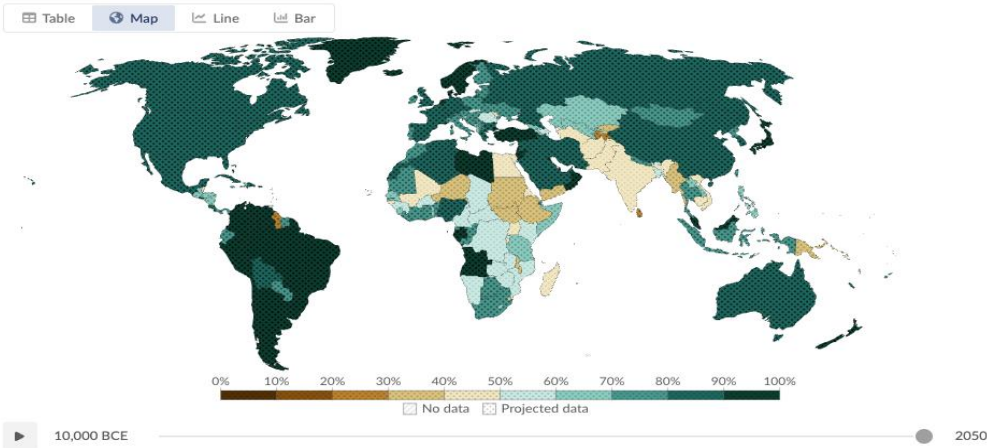
Share of the population living in urban and rural areas, combining UN estimates and projections with modelled historical data back to 10,000 BCE.



Our World
in Data

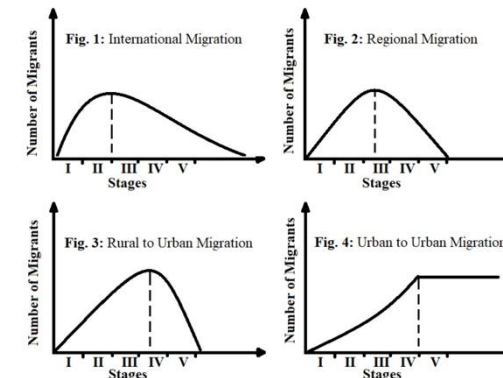
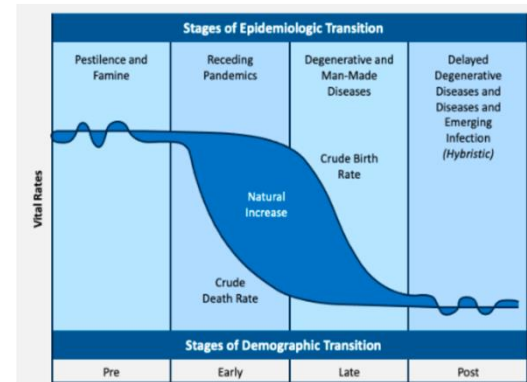
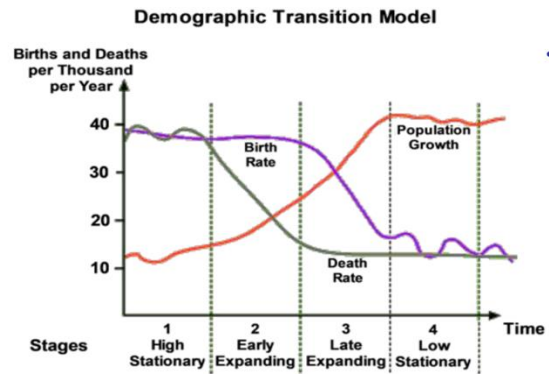
Share of population living in urban areas projected to 2050, 2050

Share of the population living in urban and rural areas, combining UN estimates and projections with modelled historical data back to 10,000 BCE.



Our World
in Data

Three Transitions



Stage	Demographic Transition	Epidemiologic Transition	Mobility Transition
I	높은 출산·사망	감염병 중심	이동 적음
II	사망 감소, 인구 급증	전염병 감소 시작	국제이주 ↑
III	출산 감소	만성질환 증가	농촌→도시 이동 ↑
IV	낮은 출산·사망	만성질환 중심	도시→도시 이동 ↑

Migration & Health

Selective migration

이주자의 건강수준이 원주민보다 좋을 수 있음

- **Healthy Migrant Effect**
 - 건강한 사람이 이주를 선택
- **Migrant Newtork effect**
 - 도착지의 커뮤니티 활용하여 자원 인프라 활용
- **Hispanic Paradox**
 - 미국 히스패닉 이민자는 낮은 소득에도 불구하고 낮은 사망률과 높은 기대수명 관찰

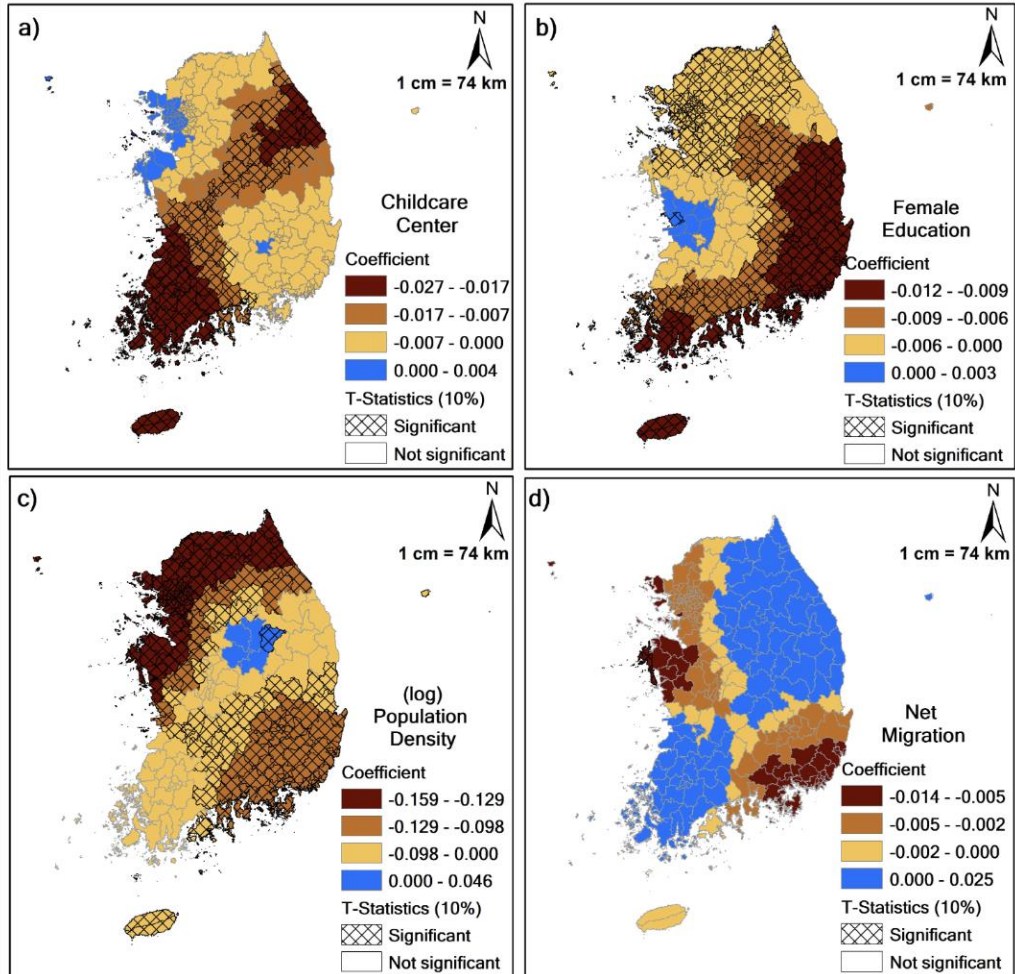
Forced migration

이주자의 건강수준이 원주민보다 좋지않음

- **취약계층의 이동**
 - 전쟁, 기후, 질병, 난민 등
- **이주과정에서 건강취약성 증가**
 - 부상 및 장애 감염병 노출
 - 영양부족 정신건강 문제 (PTSD, 우울)

Migration & Fertility

Figure 4. Spatial distribution of GWR local coefficients: (a) Childcare Centers; (b) Female Education; (c) Population Density; (d) Net Migration; (e) International Marriage; (f) Population Growth.

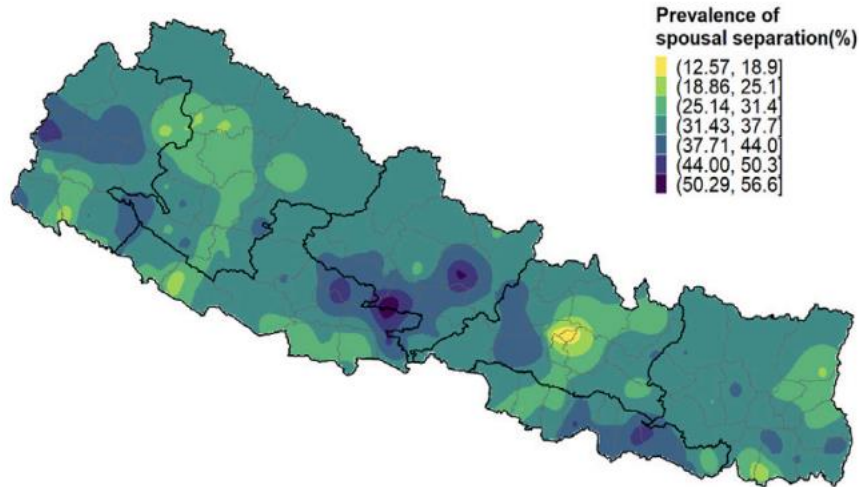


- **Adaptation Hypothesis** 도착지의 출산규범에 적응
- **Socialization Hypothesis** 출발지의 출산규범을 유지

Jung, Myunggu, et al. "Spatial variations in fertility of South Korea: A geographically weighted regression approach." *ISPRS International Journal of Geo-Information* 8.6 (2019): 262.

Migration & Reproductive Health

C



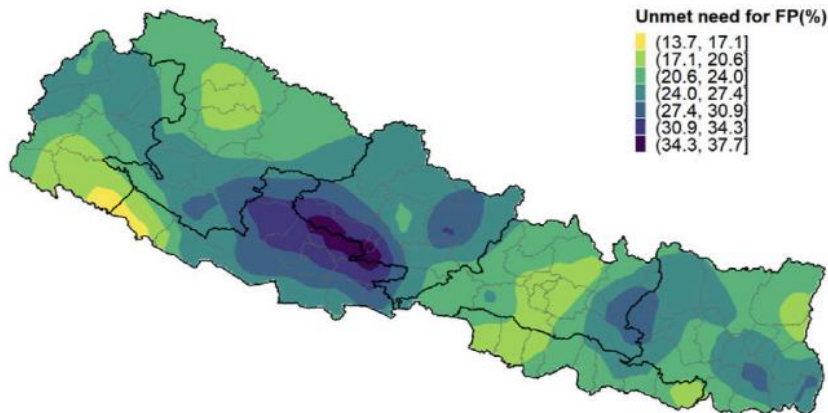
1. Reduced Access to Services

- 남편 부재로 인한 의료기관 방문 어려움

2. Uncertainty in Reproductive Planning

- 배우자 귀환 시기 불확실
- 임신 계획 수립 어려움

C



3. Gender and Decision-Making

- 배우자 부재 시
- 여성의 생식건강 의사결정 제약

Kim, Y., Hendrickson, Z. M., Shakya, M., Park, Y. S., & Jung, M. (2023). Unmet Need for Family Planning and Spousal Separation in Nepal: A Spatial and Multilevel Analysis. *International Journal of Public Health*, 68, 1606395.

감사합니다